

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Sree B	Reg. No.:	192524023
Section / Batch:	AI and DS / 2025	Date:	25/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Consider the concept map: Algorithm → Growth Function → Big-Omega → Lower Bound → Best-case Explain how lower bound represents minimum execution time. Extend the map with performance guarantee node. Justify why best-case analysis is useful. Interpret how this contributes to overall performance understanding.

Q2. Consider the concept map: Recurrence Relation → Substitution Method → Expansion → Simplification → Final Complexity Trace the step-by-step transformation of recurrence into a closed form. Extend the map with Guess and Verification nodes. Justify how substitution helps solve non-standard recurrences. Interpret how this structured approach improves understanding.

Code of Conduct
I Sree B _____ (Name / Reg No) certify that this submission is my original work and I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: Sree B

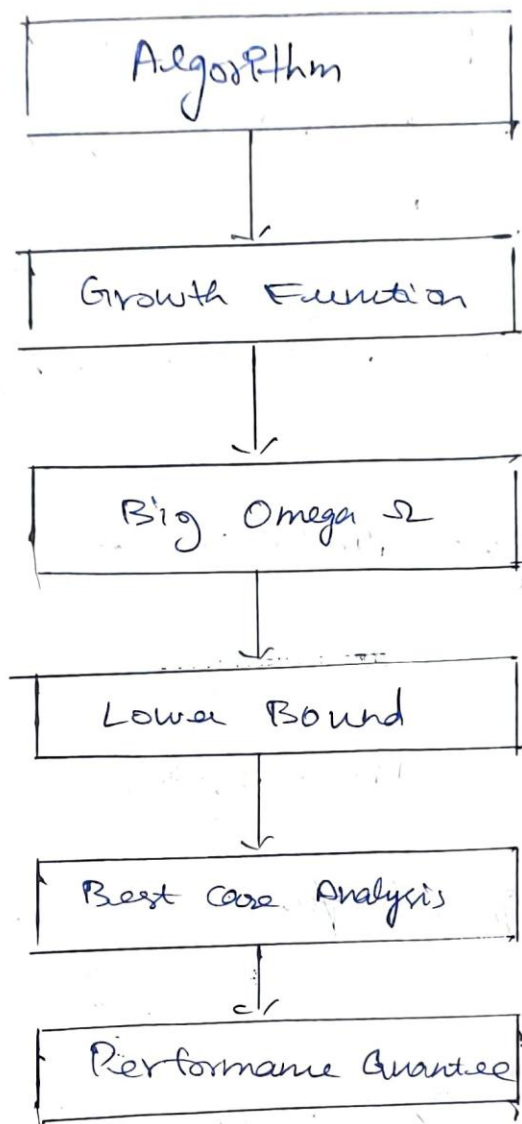
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total (Each)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



Explanation:

* Lower Bound represents the minimum execution time or algorithm requires for any valid input. It is expressed using Big Omega (Ω) notation.

* Performance Guarantee extends the concept simply by indicating that the algorithm will never perform better than the lower bound under given computational model.

* Best case analysis is useful because

(i) Show the minimum time required in the most favourable situation

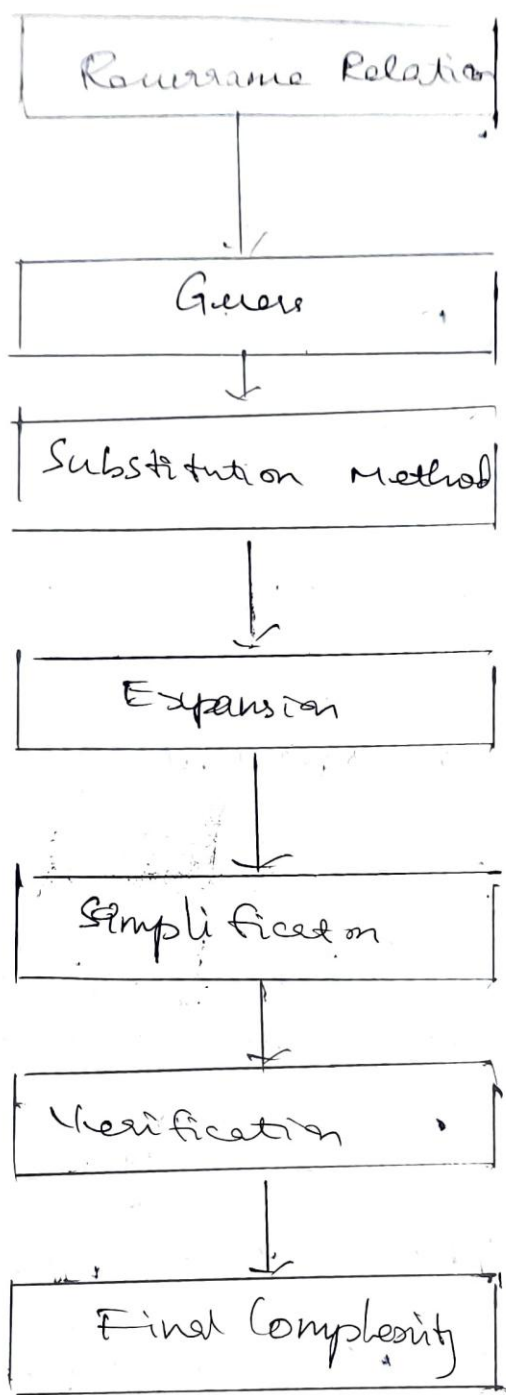
(ii) helps compare algorithms when inputs are nearly ideal.

(iii) Provide insights into the algorithm's optimal performance.

* Overall Performance Understanding.

(1) Combining growth functions, Big- Ω , lower bound and best-case analysis helps determine the minimum possible running time, complementing worst-case and average-case analyses for a complete evaluation.

②



Step by Step Transformation

- 1) Recurrence Relation $\rightarrow$ Represents the recursive running time
- 2) Guess - Assume a possible closed-form solution
- 3) Substitution Method - Substitute the guessed solution into the recurrence.
- 4) Expansion - Expand recursive terms repeatedly.
- 5) Simplification - Combine terms to obtain a compact expression.

6. Verification - check whether the guessed solution satisfies the recurrence.

7. Find Complexity - conclude the asymptotic complexity

Example $O(n \log n)$ or $O(n^2)$.

Justification:

- * Guess helps formulate a candidate solution before proving it.

- * Verification confirms that the guessed complexity is mathematically correct.

- * Substitution Method is especially useful for non-standard recurrences where direct methods like the Master Theorem cannot be applied.

- * The structured approach makes solving recurrences systematic, improves logical reasoning and ensures the derived complexity is both accurate and easy to understand.